# Procesos Estocásticos

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Catálogo

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## Continuas

**Uniforme.**
Nomenclatura:

$$X \sim U(a, b)$$

Expresiones téoricas para el rango, media y varianza.

| | |
|---|---|
| Rango | $[a, b]$ |
| Media | $\frac{a+b}{2}$ |
| Varianza | $\frac{(b-a)^2}{12}$ |

PDF:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{si } a \leq x \leq b \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 0 & \text{si } x < a \\ \frac{x-a}{b-a} & \text{si } a \leq x \leq b \\ 1 & \text{si } b < x \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \frac{b^{n+1} - a^{n+1}}{(n+1)(b-a)} \quad \Big| \quad M_X(t) = \frac{e^{tb} - e^{ta}}{t(b-a)} \quad \text{para } t \neq 0$$

**Triangular.**
Nomenclatura:

$$X \sim Tr(a, b, c)$$

Expresiones téoricas para el rango, media y varianza.

| | |
|---|---|
| Rango | $[a, c]$ |
| Media | $\frac{a+b+c}{3}$ |
| Varianza | $\frac{a^2+b^2+c^2-ab-ac-bc}{18}$ |

PDF:

$$f(x) = \begin{cases} \frac{2(x-a)}{(c-a)(b-a)} & \text{si } a \leq x \leq b \\ \frac{-2(x-c)}{(c-a)(c-b)} & \text{si } b < x \leq c \end{cases}$$

CDF:

$$F(x) = \begin{cases} \frac{(x-a)^2}{(c-a)(b-a)} & \text{si } a \leq x \leq b \\ 1 + \frac{(x-c)^2}{(c-a)(b-c)} & \text{si } b < x \leq c \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \frac{2}{b-a}\left(\frac{b^{n+2} - c^{n+2}}{(b-c)(n+1)(n+2)} - \frac{c^{n+2} - a^{n+2}}{(c-a)(n+1)(n+2)}\right)$$

$$M_X(t) = 2\frac{(b-c)e^{at} - (b-a)e^{ct} + (c-a)e^{bt}}{(b-a)(c-a)(b-c)t^2}$$

**Exponencial.**
Nomenclatura:

$$X \sim \exp(\beta)$$

Expresiones teóricas para el rango, media, y varianza:

| | |
|---|---|
| Rango | $[0, \infty)$ |
| Media | $\beta$ |
| Varianza | $\beta^2$ |

PDF:

$$f(x) = \begin{cases} \frac{1}{\beta}e^{-x/\beta} & \text{si } x \geq 0 \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 1 - e^{-x/\beta} & \text{si } x \geq 0 \\ 0 & \text{en otro caso} \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = n!\beta^n \quad \Big| \quad M_X(t) = \frac{1}{1-\beta t} \quad \text{para } t < \frac{1}{\beta}$$

**m-Erlang.**
Nomenclatura:

$$X \sim \text{Erlang}(m, \lambda)$$

Expresiones teóricas para el rango, media y varianza.

| Parámetros | $m \in \mathbb{Z}^+,\ \lambda > 0$ |
|---|---|
| Rango | $[0, \infty)$ |
| Media | $\frac{m}{\lambda}$ |
| Varianza | $\frac{m}{\lambda^2}$ |

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \frac{(m+n-1)!}{(m-1)!\,\lambda^n} \quad \Big| \quad M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^m \quad \text{para } t < \lambda$$

**Gamma.**
Nomenclatura:

$$X \sim \text{gamma}(\alpha, \beta)$$

Expresiones téoricas para el rango, media y varianza.

| Rango | $[0, \infty]$ |
|---|---|
| Media | $\alpha\beta$ |
| Varianza | $\alpha\beta^2$ |

PDF:

$$f(x) = \begin{cases} \frac{\beta^{-\alpha} x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)} & \text{si } x > 0 \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 1 - e^{-x/\beta} \sum_{j=0}^{\alpha-1} \frac{(x/\beta)^j}{j!} & \text{si } x > 0 \\ 0 & \text{en otro caso} \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \frac{\Gamma(\alpha+n)}{\Gamma(\alpha)} \beta^n \quad \Big| \quad M_X(t) = \left(\frac{1}{1-\beta t}\right)^\alpha \quad \text{para } t < \frac{1}{\beta}$$

**Weibull.**
Nomenclatura:

$$X \sim \text{Weibull}(\alpha, \beta)$$

Expresiones teóricas para el rango, media y varianza.

| Rango | $[0, \infty)$ |
|---|---|
| Media | $\frac{\beta}{\alpha} \cdot \Gamma\left(\frac{1}{\alpha}\right)$ |
| Varianza | $\frac{\beta^2}{\alpha}\left\{2\Gamma\left(\frac{2}{\alpha}\right) - \frac{1}{\alpha}\left[\Gamma\left(\frac{1}{\alpha}\right)\right]^2\right\}$ |

PDF:

$$f(x) = \begin{cases} \alpha\beta^{-\alpha} x^{\alpha-1} e^{-(x/\beta)^\alpha} & \text{si } x > 0 \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 1 - e^{-(x/\beta)^\alpha} & \text{si } x > 0 \\ 0 & \text{en otro caso} \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \beta^n \Gamma\left(1 + \frac{n}{\alpha}\right) \quad \Big| \quad M_X(t) = \sum_{n=0}^{\infty} \frac{t^n \beta^n}{n!} \Gamma\left(1 + \frac{n}{\alpha}\right)$$

**Normal.**
Nomenclatura:

$$X \sim (\mu, \sigma^2)$$

Expresiones téoricas para el rango, media y varianza.

| | |
|---|---|
| Rango | $(-\infty, \infty)$ |
| Media | $\mu$ |
| Varianza | $\sigma^2$ |

PDF:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{(2\sigma^2)}}$$

CDF:

$$F(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right] = \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(u-\mu)^2}{2\sigma^2}} du$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \mu^{n-2j} \sigma^{2j} (2j-1)!! \quad \Big| \quad M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

**Log-normal.**
Nomenclatura:

$$X \sim LN(\mu, \sigma^2)$$

Expresiones téoricas para el rango, media y varianza.

| | |
|---|---|
| Rango | $[0, \infty)$ |
| Media | $e^{\mu + \sigma^2/2}$ |
| Varianza | $e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$ |

PDF:

$$f(x) = \begin{cases} \frac{1}{x\sqrt{2\pi\sigma^2}} e^{\frac{-(\ln x - \mu)^2}{2\sigma^2}} & \text{si } x > 0 \\ 0 & \text{en otro caso} \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = e^{n\mu + \frac{1}{2}n^2\sigma^2} \quad \Big| \quad M_X(t) = \text{No existe (la integral diverge)}$$

**Beta.**

Nomenclatura:

$$X \sim \text{Beta}(\alpha_1, \alpha_2)$$

Expresiones teóricas para el rango, media y varianza.

| | |
|---|---|
| Rango | $(0,1)$ |
| Media | $\frac{\alpha_1}{\alpha_1+\alpha_2}$ |
| Varianza | $\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$ |

PDF:

$$f(x) = \begin{cases} \frac{x^{\alpha_1-1}(1-x)^{\alpha_2-1}}{B(\alpha_1,\alpha_2)} & \text{si } 0 < x < 1 \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 0 & \text{si } x \leq 0 \\ I_x(\alpha_1, \alpha_2) & \text{si } 0 < x < 1 \\ 1 & \text{si } x \geq 1 \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \frac{B(\alpha_1 + n, \alpha_2)}{B(\alpha_1, \alpha_2)} \quad \Big| \quad M_X(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \left( \frac{B(\alpha_1 + n, \alpha_2)}{B(\alpha_1, \alpha_2)} \right)$$

# Discretas

**Uniforme.**

Nomenclatura:

$$X \sim DU(i, j)$$

Expresiones téoricas para el rango, media, y varianza.

| | |
|---|---|
| Rango | $\{i, i+1, ..., j\}$ |
| Media | $\frac{i+j}{2}$ |
| Varianza | $\frac{(j-i+1)^2-1}{12}$ |

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \frac{1}{j-i+1} \sum_{x=i}^{j} x^n \quad \Big| \quad M_X(t) = \frac{e^{it} - e^{(j+1)t}}{(j-i+1)(1-e^t)} \quad \text{para } t \neq 0$$

**Bernoulli.**
Nomenclatura:

$$X \sim Bernoulli(p)$$

Expresiones teóricas para el rango, media y varianza:

| | |
|---|---|
| Rango | $\{0, 1\}$ |
| Media ($\mu$) | $p$ |
| Varianza ($\sigma^2$) | $p(1-p)$ |

PMF:

$$p(x) = \begin{cases} 1-p & \text{si } x = 0 \\ p & \text{si } x = 1 \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 0 & \text{si } x < 0 \\ 1-p & \text{si } 0 \leq x < 1 \\ 1 & \text{si } 1 \leq x \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = p \quad (n \geq 1) \quad \Big| \quad M_X(t) = (1-p) + pe^t$$

**Geométrica.**
Nomenclatura:

$$X \sim geom(p)$$

Expresiones téoricas para el rango, media y varianza.

| | |
|---|---|
| Rango | $\{0, 1, ...\}$ |
| Media | $\frac{1-p}{p}$ |
| Varianza | $\frac{1-p}{p^2}$ |

PMF:

$$p(x) = \begin{cases} p(1-p)^x & \text{si } x \in \{0, 1, ...\} \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 1-(1-p)^{[x]+1} & \text{si } x \geq 0 \\ 0 & \text{en otro caso} \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \frac{d^n}{dt^n} M_X(t)\Big|_{t=0} \quad \Big| \quad M_X(t) = \frac{p}{1-(1-p)e^t} \quad \text{para } t < -\ln(1-p)$$

**Binomial negativa.**
Nomenclatura:

$$X \sim BN(r, p)$$

Expresiones téoricas para el rango, media y varianza.

| | |
|---|---|
| Rango | $\{0, 1, ...\}$ |
| Media | $\frac{r(1-p)}{p}$ |
| Varianza | $\frac{r(1-p)}{p^2}$ |

PMF:

$$p(x) = \begin{cases} \binom{r+x-1}{x} p^r (1-p)^x & \text{si } x \in \{0, 1, \ldots\} \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} \sum_{i=0}^{\lfloor x \rfloor} \binom{r+i-1}{i} p^r (1-p)^i & \text{si } x \geq 0 \\ 0 & \text{en otro caso} \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \frac{d^n}{dt^n} M_X(t) \Big|_{t=0} \quad \Big| \quad M_X(t) = \left( \frac{p}{1-(1-p)e^t} \right)^r \quad \text{para } t < -\ln(1-p)$$

**Poisson.**
Nomenclatura:

$$X \sim Poisson(\lambda)$$

Expresiones téoricas para el rango, media y varianza.

| | |
|---|---|
| Rango | $\{0, 1, ...\}$ |
| Media | $\lambda$ |
| Varianza | $\lambda$ |

PMF:

$$p(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & \text{si } x \in \{0, 1, \ldots\} \\ 0 & \text{en otro caso} \end{cases}$$

CDF:

$$F(x) = \begin{cases} 0 & \text{si } x < 0 \\ e^{-\lambda} \sum_{i=0}^{\lfloor x \rfloor} \frac{\lambda^i}{i!} & \text{si } x \geq 0 \end{cases}$$

Momentos Analíticos y FGM:

$$\mathbb{E}[X^n] = \sum_{k=1}^{n} \left\{ {n \atop k} \right\} \lambda^k \quad \Big| \quad M_X(t) = e^{\lambda(e^t - 1)} \quad \text{para } t \in \mathbb{R}$$